Measuring Advice-Style Language in OLMo-2: Effects of Detection Rules and Response Length

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Abstract

Phrase detectors offer a simple way to compare language-model responses, but their conclusions can depend on the matching rule and the amount of text searched. We study this sensitivity in OLMo-2 using 140 prompts, public 1B and 7B checkpoints, a 120-sentence AI-label audit, and about 378,000 preference pairs per model. In the recovered final-7B responses, the advice-minus-emotional- control gap for the historical empathy marker changes from -14.7 percentage points on all archived text to $+25.3$ on the first 64 tokens. The former estimate is uncertain, and archived generations were capped at 256 tokens. The combined exact-or-embedding detector also has low precision against the audit's narrow AI labels; human validation remains pending. Under this detector, the advice-minus-facts validation gap increases from SFT to DPO while the structuring gap contracts. In exploratory paired preference-data models, broad referral/disclaimer and structuring phrases remain positively associated with selection after flexible length conditioning; narrower-referral estimates remain compatible with no association. These findings show how measurement choices affect marker comparisons, without establishing semantic empathy or causal training effects.

1 Introduction

Suppose a model answers "I've been feeling anxious and cannot sleep—what should I do?" with "That sounds really difficult. Here are a few steps you can try." The historical rules count both an empathy-opener and a structuring marker. Now consider the constructed fragment "A checklist may help. That sounds really useful for packing." It also triggers the empathy-opener rule: the phrase *that sounds really* appears in the second sentence, even though it appraises a checklist rather than a person's emotion. This is an illustration of rule behavior, not a sampled output or validation example.

Such cases matter beyond one imperfect detector. Checkpoint and condition comparisons often reduce free-form answers to occurrence rates. A longer answer offers more phrases and sentences to search, and an embedding rule can add matches that an exact rule would reject. A change in occurrence rate can therefore reflect the model response, the amount of text observed, the matching rule, or all three. Without separating them, an apparent behavioral trajectory may instead be a measurement trajectory.

This paper asks one question: *How do detection rules and response length change conclusions about advice-style language across OLMo-2 post-training?* We address three parts: **RQ1**, how detector components and text windows change measured contrasts; **RQ2**, how those contrasts vary with control condition and checkpoint; and **RQ3**, how phrase–preference associations change when response length is modeled. A **marker** is a specified textual feature, such as one of the phrases in our lexicon. A **detector** is the algorithm that decides whether a marker category occurs. A detector output becomes a **proxy** when it stands in for a broader construct such as empathy. In particular, the category named *empathy-opener* in the original artifact is searched anywhere in an answer, and *referral/disclaimer* combines referrals, disclaimers, and professional-language mentions. Neither output is a semantic judgment.

OLMo-2 is useful for this measurement study because its final Instruct model is the product of four public data-bearing stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). OLMo-2 offers an unusually complete public chain

of post-training data and intermediate checkpoints.

Findings and contributions.

1. In the recovered final-7B responses, component and output-window choices change targeted-control contrasts; an AI-label diagnostic traces every sampled union false positive to the embedding component. Human validation is pending.

2. Under the historical detector, same-interface SFT-to-DPO validation gaps increase at both model sizes while structuring gaps contract. Base-to-SFT comparisons remain confounded with prompt serialization.

3. In exploratory paired DPO models, broad referral/disclaimer and structuring associations remain positive after flexible length conditioning; both narrower-referral intervals include one.

2 Related Work

Behavioral tests and construct validity. Our targeted prompt battery follows the behavioral-testing principle that targeted capabilities require controlled tests beyond aggregate accuracy (Ribeiro et al., 2020). Public checkpoint suites support training-dynamics research (Biderman et al., 2023); we compare condition gaps across named post-training stages.

Very recent work traces situation modeling and putative mentalizing across OLMo-2 and Pythia stages while stress-testing construct validity (Rivière et al., 2026). It is closest in model family and developmental framing; its caution about mapping task outputs to cognitive constructs directly motivates our proxy terminology. Emotional-support work typically grounds support strategies in theory and human annotation (Liu et al., 2021), while empathy research warns that incompatible operational definitions undermine comparison (Lahnala et al., 2022). Recent studies compare human and LLM judgments directly (Welivita et al., 2026); in mental-health settings, LLM judges are less reliable for affective properties such as empathy (Badawi et al., 2026). Accordingly, our AI-only audit diagnoses a rule rather than validating a human construct.

Construct-validity reviews distinguish a target phenomenon from the task and score used to measure it (Bean et al., 2025). Empathy-related metric gains need not improve human preference (Lee et al., 2026), although validated LLM judgments can approach expert agreement in specific settings (Kumar et al., 2026). Our audit therefore concerns this detector and labeling task, not the reliability of all automated empathy judgments. Warmth-oriented training studies also examine accuracy and sycophancy (Ibrahim et al., 2026); our marker counts do not measure those outcomes.

Length in preference evaluation. Preference and reward-model data can encode length correlations (Singhal et al., 2023), and length-controlled evaluation can change model rankings (Dubois et al., 2024). Our contribution is narrower: we test how response length changes marker-specific exact-match differences in the OLMo-2 DPO mixtures. We report the post-hoc length strata alongside an exploratory paired conditional model whose protocol was written before its new fits, after the aggregate results were known. Both describe associations rather than a causal adjustment.

Open models, data, and the attribution boundary. OLMo-2 follows Tulu-3's public post-training recipe (Lambert et al., 2024). Its data and checkpoints permit auditable comparisons. Exact-substring search with infini-gram (Liu et al., 2024) and output-to-pretraining tracing with OLMoTrace (Liu et al., 2025) motivate our secondary corpus analysis; string co-occurrence alone does not establish attribution.

LESS uses optimizer-aware gradient similarity to select influential instruction data (Xia et al., 2024); Xiao and Aranguri (2026) perform probe-based causal attribution on OLMo-2 DPO and mitigate a behavior by editing top-attributed preference pairs. Our evidence is observational: it identifies detector changes and phrase co-occurrence across public stages, but does not establish which examples caused an output.

3 Measurement and Study Design

3.1 Operational categories and fixed detector

A hand-seeded, inspectable lexicon specifies four marker categories using the artifact's historical names: *empathy-opener*, *validation-phrase*, *referral/disclaimer* (artifact key `disclaimer`), and *structuring-marker* (`structure`). The names label rules rather than validated constructs. The empathy-opener rule searches every sentence, not only an opening span. Referral/disclaimer

is a broad combined rule that also includes professional-language fragments such as "consult a" and "medical professional"; it is not a detector of referrals alone. A fifth historical category, `crisis_referral`, is excluded from the four-category analysis after producing 0 true positives and 11 false positives in the audit.

The historical comparisons use archived detector outputs and the original primary threshold rather than rules selected for the present revision. The detector takes the union of case-insensitive substring matching and an e5-small embedding rule (Wang et al., 2024): a response is positive when either an exact phrase appears or any sentence reaches cosine similarity $\tau = 0.82$ to a category exemplar. We also examine $\tau = 0.80$ and 0.84. Longer responses supply more sentences and therefore more opportunities for the embedding rule to fire. The complete specification used to reproduce the July final-7B response analysis is in Appendix B; the raw-cache diagnostics preserve that version rather than retrospectively improving it. Archived stage summaries are analyzed as historical outputs. Because most stagewise raw responses are unavailable, we do not claim to have re-executed every stage under a byte-identical detector.

Construction and pilot overlap. The accompanying pilot file contains eight personal-advice prompts and four general-facts controls used while hand-seeding the proxy; all 12 reappear in the 140-prompt battery. We therefore do not call the full battery held out. A final-7B sensitivity excluding those 12 leaves the advice–factual gaps similar (Appendix D). The sample sizes refer to distinct datasets: the 140 prompts form the marker evaluation, whereas 120 stratified final-model sentences form an AI-label diagnostic. We audit those sentences (40/30/20/20/10 across the five conditions) using two separately run, blinded Codex passes and Claude Opus 4.8 adjudication of their three disagreements. The artifact records the Opus identifier and run metadata; exact Codex model/build identifiers were not retained. This is an AI diagnostic, not human validation. The revision materials include the labels, adjudication reasons, and detector (Section 4.1; Appendix B).

3.2 Prompt conditions and generation

Five conditions target different aspects of advice-style responses: *personal advice* (60 personal or emotional help requests), *general facts* (30 knowledge questions), *emotion without a request* (20 prompts about emotional topics without asking for help), *neutral how-to* (20 requests for practical advice without emotional stakes), and *health facts* (10 impersonal clinical-information questions). These controls separate request type, emotional content, and domain. They target conceptual axes but are not paired counterfactuals or topic-matched scenarios; non-significance is not interpreted as equivalence. The final-model battery and the canonical personal-advice/general-facts checkpoint trajectories use five sampled generations per prompt (temperature 0.7, top-p 0.9, 256 new tokens, distinct seeds). The Base/SFT/DPO five-condition extension uses four; its balanced comparisons use seeds 0–3 for every condition.

Table 1 separates the study's evidence sources. The final condition battery asks whether a marker is specific to personal advice. The checkpoint grid asks when advice-minus-general-facts gaps change. The AI-label diagnostic asks how the detector relates to narrow sentence labels. The DPO analysis asks whether exact phrases differ between chosen and rejected answers, and how those differences relate to response length.

3.3 Inference at the prompt level

The clustering unit is the prompt: each prompt contributes its mean detector-positive indicator over its four or five generations, and condition rates are means over prompts. Historical between-condition inference uses prompt-clustered permutation tests (10,000 permutations of prompt-level means; two-sided; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini–Hochberg correction is applied separately to the 24 final-condition contrasts, 28 advice/general-facts adjacent-stage interactions, 24 targeted-control stage interactions, and 8 historical DPO pairwise tests. We do *not* treat generations as independent Bernoulli trials: all condition and stage tests operate on prompt-level generation means.

For a new retrospective detector-component check, we directly estimate the change in each advice-minus-control gap when replacing union with exact-only matching on the same archived text. We resample prompts within condition 20,000 times, retaining all five generations and both detector outputs together. Shared draws yield approximate simultaneous 95% intervals across the four changes using the maximum absolute standardized bootstrap deviation. This exploratory analysis is

Analysis	Material	Comparison	Main scope limit
RQ1: Detector/window	700 cached final-7B responses; 140 prompts, five generations	Exact-only versus union contrasts; all archived text versus 64/128-token prefixes	Generations capped at 256 tokens; retrospective and restricted to final 7B
RQ1: AI-label diagnostic	120 sentences from 97 final responses	Historical detector versus narrow AI consensus labels	No human labels; referral target is narrower than the detector
RQ2: Final conditions	140 prompts, five conditions, five generations	Personal advice versus four controls at final 1B and 7B	Author-written controls are not topic-paired counterfactuals
RQ2: Checkpoint grid	90 personal-advice/general-facts prompts, five generations	Changes in the advice-minus-facts gap across public 1B and 7B stages	Base-to-SFT also changes prompt serialization; most raw responses are unavailable
RQ3: DPO pairs	378,301 1B and 378,341 7B pairs	Exact chosen-versus-rejected matches, with paired flexible length conditioning	Observational; conditioning changes the estimand; mixtures share 21,508 pairs

Table 1: Evidence map. Each analysis answers a different part of the measurement question and has a different unit and scope limit.

conditional on the historical prompt frame; it does not supply human validation or a fresh evaluation.

3.4 Exploratory paired DPO length model

For each of three exact marker groups and each model size, the new paired model uses the conditional likelihood $\sum_i \log \sigma[\boldsymbol{\beta}^\top(\mathbf{x}_{ic} - \mathbf{x}_{ir})]$ with no intercept. The features are the chosen-minus-rejected marker flag and differences in a restricted cubic spline of $\log(1 + \text{words})$. Five pooled knots were fixed at the 5th, 27.5th, 50th, 72.5th, and 95th percentiles (7, 78, 204, 374, and 739 words) before fitting. Intervals use a prompt-cluster sandwich covariance with finite-cluster correction. The resulting conditional odds ratio (OR) equals one when the model estimates no marker–choice association at the modeled lengths; it is not a percentage-point change in choice probability. Conditioning on an observed response property changes the estimand and does not remove bias or identify a causal effect. The six fits are exploratory, not preregistered or confirmatory; Benjamini–Hochberg values across them are secondary summaries.

3.5 Checkpoint evaluation

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct; 7B Base→SFT→DPO→Instruct. Base checkpoints have no chat template and use a plain Question/Answer format, whereas post-trained checkpoints use their official chat template. Thus Base→SFT is a stage-plus-interface transition and cannot isolate SFT training. No serialization crossover was run; we demote that transition throughout. The stagewise grid runs the 90 personal-advice plus general-facts prompts at every checkpoint. All five conditions run on the final models and, in the four-seed specificity extension, at Base, SFT, and DPO.

Corpus searches provide secondary evidence of phrase co-occurrence, not causal attribution. Appendix G details their data, search scope, length-matched controls, and missing Stage-2 index.

Reproducibility. Stagewise, condition, and corpus numbers derive from `summary_v3.json` and accompanying scripts under `experiments/`. The revision materials include prompts, the lexicon, seeds, thresholds, and adjusted-p output. Audit numbers are materialized in `ai_consensus_validation_v3.json`, which also contains both 120-item passes, consensus labels, Opus decisions, and reasons. The paired DPO features, parity manifest, conditional estimates, and dated exploratory protocol are stored under `output/revision/dpo_conditional/` and `out/paper/`. The retrospective component contrasts are reproduced by `experiments/window_component_reanalysis_20261109.py`; estimates and input hashes are in `output/review_2026-11-09/component_reanalysis/paired_measurement_contrasts.json`.

4 Results

We first audit the measurement rule, then report condition and checkpoint comparisons under that rule, and finally examine the association between DPO phrase differences and response length. This order matters because the audit limits what the subsequent detector-positive rates can mean.

Marker	Control subtracted from advice	Exact, all text	Union, all text	Union, 64	Union, 128
empathy	Emotion, no request	.323	−.147	.253	.063
validation	Emotion, no request	.153	.250	.177	.210
referral/disclaimer	Health facts	.157	.090	.113	.117
structuring	Neutral how-to	.390	−.030	−.033	−.030

Table 2: RQ1: detector and text-window sensitivity in the 700 recovered final-7B responses. Each cell is the personal-advice rate minus the named control rate; multiplying by 100 gives percentage points. "All text" uses the complete archived generation, capped at 256 new tokens; 64 and 128 use token prefixes. Union combines exact and embedding matches at $\tau = .82$. These exploratory estimates describe measurement sensitivity; separate tests at each window do not by themselves test the change between windows.

4.1 RQ1: Detector and text-window sensitivity

A condition-stratified random audit samples 120 final-model sentences across all five conditions (40 personal advice, 30 general facts, 20 emotion without a request, 20 neutral how-to, and 10 health facts). The sample contains 56 sentences from the 1B model and 64 from the 7B model, drawn from 97 distinct seed-0 responses. It was not stratified on detector predictions, and multiple sentences can come from one response. Two independent blinded Codex passes agree exactly on 117 of 120 sentences (97.5%). Category agreement is 99.2–100%; Cohen's κ is .885 for empathy expression, .663 for validation, 1.000 for referral, and .978 for structuring. Claude Opus 4.8 adjudicates the three disagreements. These are AI-consensus labels under specific prompts and models, not human ground truth.

Against that consensus, the union detector has low precision and category-dependent recall (Table 3). The audit's referral label is narrower than the detector: it requires an actual recommendation to seek a qualified professional, while the historical referral/disclaimer rule also matches disclaimers and professional-language mentions. Its 0.06 precision therefore reflects both broad detector behavior and target misalignment; it should not be read as the accuracy of a pure referral detector. Empathy and validation recall are each 0.50. Errors vary by condition, so subtracting condition rates does not guarantee cancellation. The detector overpredicts positives in this sample, although the rates are not formal upper bounds because recall is below one.

Exact substring matching has no false positives in these 120 sentences but misses many AI-labeled positives. Its true-positive/false-positive/false-negative counts are 2/0/2 for empathy expression, 1/0/1 for validation, 1/0/0 for referral, and 9/0/21 for structuring. Every union false positive is introduced by the embedding component. The union raises structuring recall from 0.30 to 0.87, with lower precision. Component choice is therefore part of the measurement question.

The recovered 700-response cache permits a separate response-level diagnostic for the final 7B checkpoint only. Advice responses average 244 tokens and factual responses 187; the three targeted-control means are 251–255 tokens. Table 2 compares the matching rules and text windows. All archived generations were capped at 256 new tokens: "all text" means the complete cached text, not necessarily a completed answer. For the category historically called empathy-opener, union rates on all text are .6133 for advice and .7600 for emotion without a request (a gap of -14.7 percentage points, $p = .08839$). At 64 tokens they are .3833 and .1300 (a gap of $+25.3$ points, $p = .00340$); at 128 tokens they are .4933 and .4300 ($+6.3$ points, $p = .48845$). The point estimate changes sign at 64 tokens; the all-text and 128-token comparisons are uncertain. These window checks are exploratory, and their reported p-values are unadjusted. Validation and referral/disclaimer contrasts remain positive across the tested windows, while structuring remains near zero. One recovered health-facts response illustrates the opportunity mechanism: "Here are some key points based on research findings" occurs after the first 64 tokens, so the exact structuring rule fires on the cached text but not on its 64-token prefix. This is a detector example, not a judgment about the answer's quality. These component and window checks qualify the final-7B result; raw responses are unavailable for the other checkpoints, so they do not validate the stagewise trajectories.

A paired component analysis directly tests changes in the contrasts. Replacing union with exact-only matching changes the empathy gap by $+47.0$ percentage points (simultaneous 95% inter-

category	P	R	F1	tp/fp/fn
empathy expr.	0.22	0.50	0.31	2/7/2
validation	0.33	0.50	0.40	1/2/1
ref./discl.	0.06	1.00	0.12	1/15/0
structuring	0.49	0.87	0.63	26/27/4

Table 3: Union-detector precision, recall, and F1 against the adjudicated 120-sentence AI consensus ($\tau = 0.82$). This is a diagnostic against an AI annotation construct, not human validation. The referral annotation is narrower than the combined historical referral/disclaimer rule. Response-cluster bootstrap intervals appear in Appendix I.

val $[28.1, 65.9]$) and the structuring gap by $+42.0$ points ($[21.9, 62.1]$). The validation change is -9.7 points ($[-22.7, 3.3]$), and referral/disclaimer changes by $+6.7$ points ($[-14.7, 28.0]$). Thus detector choice changes the empathy and structuring contrasts within this archive. These intervals cover the four component changes jointly; the window comparisons above remain descriptive exploratory checks.

4.2 RQ2: Controls and checkpoint comparisons

Table 4 gives mean prompt-level proxy-positive rates per condition on the final models; Table 5 tests advice against each targeted control. Against general-facts controls every category separates sharply in both models (all prompt-clustered $p \leq .0002$). For referral/disclaimer, the 1B model has rates .900 on personal advice and .080 on general-facts prompts; the 7B model has rates .913 and .060.

Targeted controls change the interpretation. At $\tau = .82$, the category historically called empathy-opener fires more often when emotion is present without an advice request than advice: .790 versus .590 for 1B ($p = .010$), and .760 versus .617 for 7B ($p = .098$). These are archived summary values. In the separately reconstructed final-7B cache, the corresponding all-text advice rate is .6133 ($p = .08839$ for the $-.1467$ gap), and the point estimate changes sign at 64 tokens as reported above. Structuring is high on both advice forms: the personal-minus- neutral-how-to difference is .000 for 1B ($p = 1.0$) and $-.030$ for 7B ($p = .574$). These are tests for a difference, not equivalence tests.

The broad referral/disclaimer category is also common outside personal advice. It fires on .740 of neutral-advice prompts for 1B and .720 for 7B, and on .820 of health-facts prompts for both models. Its advice-minus-health-facts difference is small and not significant: .080 for 1B ($p = .352$) and .093 for 7B ($p = .159$). The validation-phrase union output is higher on advice than every targeted control for both models (all unadjusted $p \leq .0044$; all BH-adjusted $q \leq .0082$). These marginal profiles differ, but weak correlations do not establish the absence of a latent response template. The comparisons are detector-dependent, and the small, mixed 1B–7B differences do not support a scaling claim.

Marker co-occurrence. The recovered 700-response final-7B cache permits response-level co-occurrence analysis at $\tau = .82$. Among advice responses, 4.7% contain one proxy, 29.0% two, 40.7% three, and 25.7% all four (none contain zero). Referral/disclaimer is present in 93.0% of validation-positive advice responses, while pairwise phi correlations range from $-.14$ to .21. The categories often co-occur, but these weak correlations alone neither establish nor rule out a latent response template. The accompanying materials include condition-level joint distributions and all pairwise correlations (Appendix D).

Checkpoint gaps change by marker. Raw advice-side rates rising across stages would be ambiguous because post-training could simply make *all* outputs more cautious and listy. A more informative operational quantity is the personal-advice-minus-general-facts gap Δ_s at each stage (Table 10) and its *change* across adjacent stages, a stage$\times$condition difference-in-differences tested with a prompt-clustered permutation test on per-prompt stage trajectories (Table 7; 10k permutations, with 60 personal-advice prompts and 30 general-facts prompts).

All eight Base-to-SFT/interface changes are positive and six survive BH correction. For referral/disclaimer, the change is +.253 for the 1B model and +.243 for the 7B model; for structuring it is +.337 and +.227, respectively. Validation does not survive correction ($q = .137$ for 1B and $q = .121$ for 7B). Every category is already detector-positive at Base. Because Base uses plain QA and SFT uses the chat template, these changes cannot isolate training from serialization.

The same-interface SFT-to-DPO transition differs by category. The validation gap increases by +.213 for 1B ($q = .002$) and +.147 for 7B ($q = .021$). The 1B referral/disclaimer gap increases by +.190 ($q = .002$), while its 7B change of

Marker output	Model	Personal advice	General facts	Emotion, no request	Neutral how-to	Health facts
empathy opener	1B	0.590	0.040	**0.790**	0.020	0.340
	7B	0.617	0.027	**0.760**	0.070	0.360
validation phr.	1B	0.363	0.007	0.120	0.000	0.040
	7B	0.383	0.013	0.130	0.030	0.060
ref./discl.	1B	0.900	0.080	0.510	0.740	0.820
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.580	0.820	**1.000**	0.920
	7B	0.970	0.547	0.830	**1.000**	0.840

Table 4: Mean prompt-level union-detector rate per condition (final models, $\tau = 0.82$; each prompt averaged over 5 generations). Bold marks the cells that break the "advice-specific" reading: the empathy-opener output peaks when emotion appears without an advice request, and structuring is at ceiling on neutral how-to prompts. The audit shows overprediction in its sentence sample but does not recalibrate these response-level rates (Section 4.1); `crisis_referral` is excluded (0 TP / 11 FP in the 120-sentence audit).

Marker	Model	Emotion, no request	Neutral how-to	Health facts
empathy	1B	−.200 (.010)	.570 (.0002)	.250 (.016)
	7B	−.143 (.098)	.547 (.0002)	.257 (.031)
validation	1B	.243 (.003)	.363 (.0002)	.323 (.003)
	7B	.253 (.003)	.353 (.0003)	.323 (.004)
referral	1B	.390 (.0002)	.160 (.006)	.080 (.352)
	7B	.373 (.0002)	.193 (.001)	.093 (.159)
structuring	1B	.180 (.0002)	.000 (1.0)	.080 (.018)
	7B	.140 (.004)	−.030 (.574)	.130 (.026)

Table 5: Each cell is personal advice minus the named targeted control (prompt-clustered permutation p, 10k draws). Benjamini–Hochberg correction across the 24 displayed tests leaves the validation-phrase result unchanged (all $q \leq .0082$). Non-significant cells mean "no detected difference," not equivalence.

+.083 is not significant after correction ($q = .203$). The empathy-opener changes are also not significant after correction. The structuring gap contracts by $-.367$ for 1B and $-.277$ for 7B (both $q = .002$), because DPO raises detector-positive structuring on factual answers while advice-side rates are near ceiling. Preference-data co-occurrence cannot independently explain these trajectories: length-stratified estimates differ sharply, while the exploratory conditional models retain smaller broad-referral and structuring associations (Table 6).

No RLVR or final-merge gap change is significant (all raw $p \geq .168$). These tests describe the outputs of the fixed detector. Because the audit finds condition-dependent false positives and does not span checkpoints, they do not by themselves establish changes in human-judged behavior prevalence.

Comparisons with targeted controls. The four-seed Base/SFT/DPO extension tests 24 planned stage×control interactions (Appendix F). Four survive BH correction at $\tau = .82$. For empathy relative to emotion without an advice request, the 7B Base→SFT/interface transition increases the advice gap ($+0.358$, $q = .016$), but DPO reverses it at both 1B (-0.467, $q = .001$) and 7B (-0.642, $q = .001$). At 7B, SFT also shifts structuring toward neutral advice (-0.208, $q = .026$). No validation or referral interaction survives $q < .05$; 1B DPO validation tendencies have $q = .059$–$.095$. The threshold sweep yields three significant interactions at $\tau = .80$, four at .82, and five at .84; only the two 7B empathy transitions survive at all three thresholds. Thus the broad advice-minus-general-facts changes cannot all be read as advice-specific.

4.3 RQ3: Phrase preference and response length

The exact mixtures contain 378,301 preference pairs for 1B and 378,341 for 7B. Chosen responses are slightly shorter in aggregate, yet exact referral/disclaimer and structuring matches favor the chosen side (Table 6). Only 19.1% and 18.7% of pairs fall within ± 20 words. In that post-hoc stratum, broad-referral differences are near zero and structuring differences remain positive but much smaller; outside it, differences follow the longer response side. Full paired risk differences, uncertainty, and length bands appear in Appendix H.

The conditional analysis was specified before its new fits but after the aggregate results, so it is exploratory. After flexible length conditioning, broad referral/disclaimer and structuring remain positively associated with the chosen side at both sizes (Table 6); both narrower-referral intervals include one. OR $= 1$ means no modeled marker–choice association, not a change of zero in choice probability. One-pair-per-prompt

Phrase group	1B conditional OR [95% CI]	7B conditional OR [95% CI]
referral/disclaimer	1.088 [1.026, 1.154]	1.134 [1.067, 1.205]
narrower referral	1.095 [0.972, 1.234]	1.121 [0.996, 1.263]
structuring	1.128 [1.108, 1.149]	1.190 [1.168, 1.213]

Table 6: Exploratory associations in the 1B/7B DPO training mixtures. The paired conditional OR flexibly models both response lengths; intervals use prompt-cluster sandwich covariance. Marker-only estimates are in Appendix H.

Model	Marker	Base→SFT + interface	SFT→DPO
1B	empathy opener	+0.190 (.004)	+0.117 (.121)
1B	validation phrase	+0.093 (.137)	+0.213 (.002)
1B	referral/disclaimer	+0.253 (.002)	+0.190 (.002)
1B	structuring	+0.337 (.002)	−0.367 (.002)
7B	empathy opener	+0.263 (.002)	+0.070 (.387)
7B	validation phrase	+0.103 (.121)	+0.147 (.021)
7B	referral/disclaimer	+0.243 (.002)	+0.083 (.203)
7B	structuring	+0.227 (.002)	−0.277 (.002)

Table 7: Change in the personal-advice-minus-general-facts detector gap across transitions (plus-one-corrected prompt-clustered permutation tests, 10k draws; displayed q is BH-adjusted across all 28 adjacent-stage×proxy tests). The Base→SFT column also changes the input serialization and is not a clean SFT effect. Later raw $p \geq .168$.

and leave-largest-source-out checks preserve the broad-referral and structuring directions. The mixtures share 21,508 content-identical pairs, limiting independent-replication claims. Source heterogeneity, the 19,890 retained 7B null-source rows, and the forest plot are reported in Appendix H.

5 Discussion

Which conclusions survive the checks. The final-7B empathy point estimate changes sign with the output window, and the structuring contrast changes between exact and embedding-inclusive rules. Validation remains above its strongest targeted control at all three tested thresholds, while no RLVR or final-merge gap change is significant. These are detector-defined facts; stability across three thresholds does not establish construct validity.

Length and preference data. Longer responses have more opportunities to contain an exact phrase. Aggregate DPO differences largely follow the longer side, yet exploratory paired models retain positive broad-referral and structuring associations after flexible length conditioning; narrower-referral intervals include one. These conditional associations do not identify causal effects.

Implications for measurement. Broad category names should not be mistaken for narrow constructs. Corpus matches show where strings occur, not which examples caused an output. Future work should align rules with independent human labels and cross checkpoint with prompt interface before making semantic or training-stage claims.

6 Conclusion

This OLMo-2 case study shows that detector components, text windows, and control conditions can change the apparent profile of advice-style markers. Preference- data associations also depend on the phrase definition and treatment of length. Reporting these choices makes marker comparisons easier to audit. Independent human labels and fresh checkpoint responses are still needed to establish whether the observed patterns describe semantic behavior beyond these rules and archived outputs.

7 Limitations

All analyses are observational. Checkpoint and corpus comparisons do not attribute outputs to training examples. Base uses plain QA while later checkpoints use chat serialization, confounding every Base-to-SFT contrast.

The 120-sentence audit uses AI-consensus labels and has no independent human labels. It does not validate response-level prevalence, samples 97 seed-0 responses, and has wide category intervals. Raw-window checks cover only final 7B and cannot validate other checkpoint trajectories.

The targeted controls are not topic-paired to the advice prompts, and a non-significant contrast does not show equivalence. Twelve pilot prompts were reused; excluding them changes final-7B advice-minus-general-facts gaps by at most .045. The small seed sets do not remove sampling uncertainty.

The DPO length bins are post hoc. The conditional model's dated protocol was written before its new fits but after those aggregate results were known, so the analysis is exploratory rather than preregistered or confirmatory. Conditioning on re-

sponse length changes the estimand, and an OR is not a probability change. The mixtures share 21,508 content-identical pairs. SFT recoverability uses only three to six distinct three-word types. Stage-2 Dolmino-Mix-1124 lacks a public index, and Stage-1 novelty is verbatim-only and length-confounded. One model family, two sizes, one decoding configuration, and modest author-written prompt sets support no scaling or cross-family claim.

Ethics Statement

Data and human subjects. We use public corpora and open-weight OLMo-2 checkpoints. We did not recruit participants or collect new human annotations for the analyses reported here. The 140 prompts are author-written hypothetical situations without personal information.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns surface phrases that can mark safety-relevant moves such as professional referrals. Within the AI-labeled audit sample, the detector produces many false positives; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety-relevant marker inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and provide an analysis tool, not any model or dataset that could cause direct harm.

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A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top-p 0.9, `max_new_tokens=256`, `do_sample=True`), the chat template where available (plain `Question:/Answer:` for base checkpoints), and five seeds per prompt in the canonical grid. The battery: 60 advice, 30 factual, 20 emotion-without-request prompts, 20 neutral how-to prompts, and 10 health-facts prompts; the complete machine-readable list ships as `experiments/prompts_full.json`. The Base/SFT/DPO five-condition extension uses four seeds; all balanced extension tests use seeds 0–3 for every condition. Eight advice and four general-facts prompts were reused from the earlier 12-prompt pilot; the other 128 are new to the full battery.

Examples.

- Advice: `I've been feeling really anxious lately and can't sleep. Any advice?`
- Factual: `How does photosynthesis work?`
- Emotional non-advice: `Describe the common symptoms of loneliness.`
- Neutral advice: `How should I organize my bookshelf?`
- Domain factual: `What are common clinical treatments for insomnia?`

B Surface-marker lexicon and detector threshold

This section records the executable four-category detector in `experiments/raw_output_robustness.py`, used to reproduce the July final-7B response analysis. Historical notebook variants also included a crisis category, which is excluded from the reported four-category analysis

after its audit failure. A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (`intfloat/multilingual-e5-small`, `query:`/`passage:` prefixes), with $\tau = 0.82$ primary. Every sentence is eligible; despite its historical name, `empathy_opener` is not restricted to an opening span.

Lexicon (substring match).

- **empathy-opener proxy** (`empathy_opener`): `i'm sorry to hear`, `i am sorry to hear`, `i'm really sorry`, `that sounds really`, `that sounds very`, `i can understand`, `i can imagine`, `it's understandable`, `that must be`, `i'm here for you`, `i hear you`, `it sounds like you`
- **validation-phrase proxy**: `your feelings are valid`, `it's okay to feel`, `it is okay to feel`, `it's normal to feel`, `it's completely normal`, `what you're feeling`, `there's nothing wrong with`, `it's natural to`
- **referral/disclaimer proxy** (`disclaimer`): `consult a`, `talk to a`, `speak with a`, `see a doctor`, `seek medical`, `i am not a doctor`, `i'm not a doctor`, `i'm not a medical`, `medical professional`, `healthcare professional`, `mental health professional`, `seek professional`, `professional help`
- **structuring-marker proxy** (`structure`): `here are some`, `here are a few`, `consider the following`, `steps you can take`, `a few things you can`, `first,`, `secondly,`

e5 seed exemplars. Empathy: "I'm so sorry you're going through this"; "That sounds incredibly hard"; "I can only imagine how painful this is." Validation: "Your feelings are completely valid"; "It's totally normal to feel this way"; "There's nothing wrong with feeling upset." Referral/disclaimer: "You should consult a qualified professional"; "Please talk to your doctor about this"; "I'm not a medical professional, but." Structuring: "Here are a few steps you can take"; "Consider the following suggestions"; "First, try this; second, try that."

C Threshold sweep

Table 8 reports the exact contrasts at $\tau = 0.80$, 0.82, and 0.84. Structuring is stable: advice and neutral-advice remain within .05 in every cell. Validation exceeds its strongest control in all six cells. Empathy's emotional-minus-advice point estimate changes sign at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was fixed before the 140-prompt battery, but its qualitative pattern should not be generalized across detector thresholds.

Model	Contrast	.80	.82	.84
1B	empathy: emot.−adv.	.030	.200	−.107
7B		.030	.143	−.280
1B	validation: adv.−max ctrl.	.030	.243	.107
7B		.047	.253	.170
1B	referral: adv.−domain	−.003	.080	.093
7B		.017	.093	.280
1B	structuring: adv.−neutral	.000	.000	.047
7B		.000	−.030	−.010

Table 8: Threshold robustness of the four targeted-control contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on personal advice and neutral how-to.

D Final-7B response-level robustness

The recovered `final_7b_answers.jsonl` contains all 700 final-7B responses (140 prompts $\times$ five seeds). Table 9 shows the opportunity imbalance. "Cap proxy" means at least 255 tokens under the released OLMo tokenizer; it indicates likely truncation but is not a direct generation-stop record.

Condition	n	mean tok.	mean sent.	cap proxy
advice	300	243.8	13.5	.880
factual	150	187.4	7.8	.573
emotional	100	253.5	10.9	.980
neutral how-to	100	254.6	13.8	.970
health facts	50	250.5	10.2	.960

Table 9: Final-7B response length and generation-cap proxy.

Table 2 in the main text recomputes each targeted-control contrast using exact substring

matching alone and the exact-or-embedding union, over all archived text and fixed prefix windows. The recovered-cache all-text union empathy rate is .6133 on advice, compared with .617 in the older summary. This is a discrepancy of approximately one positive response among 300, not a rounding difference; we preserve the two evidence sources separately. The control rate is .7600. The gap's point estimate changes sign at 64 tokens. The all-text and 128-token contrasts remain uncertain, so these are not two established opposite effects. Exact-only matching instead makes structuring look advice-specific relative to neutral how-to; the embedding component removes that contrast.

Among advice responses, union detections include one category in 4.7% of cases, two in 29.0%, three in 40.7%, and four in 25.7%. Pairwise phi correlations range from $-.14$ to .21; referral/disclaimer appears in 93.0% of validation-positive responses. Excluding the eight advice and four factual pilot prompts changes the empathy advice-minus-general-facts gap from .587 to .542, validation from .367 to .369, referral/disclaimer from .850 to .831, and structuring from .423 to .458.

E Stagewise rates under the historical detector

Table 10 supplies the rates underlying the transition results in Section 4.2. Parentheses contain the advice-minus-general- facts gap. The rates are historical detector outputs, not human judgments.

F Checkpoint interactions with targeted controls

Table 11 reports every planned interaction in the four-seed Base/SFT/DPO extension. Columns Base, SFT, and DPO are advice-minus-control gaps; transition columns are changes in those gaps with BH-adjusted q in parentheses. The family contains all 24 displayed transition tests.

G Per-phrase recoverability and leave-one-out

Corpus searches provide secondary context. For each stage's available indexed data we search the exact Stage-1 corpus (`v4_olmo-mix-1124_llama`; novelty = 1−verbatim coverage), each model's exact SFT mixture (`tulu-3-sft-olmo-2-mixture-0225` for 1B; `tulu-3-sft-olmo-2-mixture` for 7B) in DuckDB with assistant-only scoping (866,138 conversations for 1B and 939,344 for 7B; oasst1 (Köpf et al., 2023) is not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares seeded marker phrases against length-matched topical controls from the same answers, per-million documents, with a distinct-phrase leave-one-out. Stage-2 mid-training uses Dolmino-Mix-1124, for which no public index was available; the corpus trace is therefore incomplete.

Per-phrase assistant-only counts against each model's exact mixture (`-0225`, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

Length-matched buckets (occurrence-weighted). Seeded-marker vs topical per-million for 1B: 2-word 1884.3 vs 1760.2 (1.1×, n_{occ}=14), 3-word 12900.2 vs 629.8 (20.5×, 45), 4-word 1596.9 vs 11.7 (136.9×, 30). For 7B, 2-word 1784.5 vs 996.8 (1.8×, 20), 3-word 9354.4 vs 1280.4 (7.3×, 55), 4-word 1396.6 vs 96.2 (14.5×, 24). **Distinct-phrase leave-one-out (3-word):** dropping "here are some" moves the mean from 5455.2 to 1305.2/M at 1B (3 types) and from 3026.0 to 1065.9/M at 7B (6 types).

H DPO phrase and source robustness

Table 13 gives paired risk-difference and matched-odds-ratio intervals. The historically named "high-precision" referral set requires an explicit doctor or professional object (e.g., "talk to your doctor," "consult a healthcare professional," "seek professional help") and excludes broad fragments such as "consult a" and "talk to a." Because its semantic precision has not been human-validated, we call it the narrower referral-phrase set. Intervals use the discordant-pair normal approximation for the risk difference and log-MOR; the accompanying scripts also report exact McNemar p-values.

Table 14 is the requested within-pair length diagnostic. The near-antisymmetric signs in the longer-response strata show that the aggregate chosen-side effect is strongly associated with relative response length. The comparable-length band is descriptive and post hoc; it is not a substitute for the separately reported exploratory conditional model.

Model	Stage	Empathy	Validation	Referral/disclaimer	Structuring
1B	Base	.217 (.204)	.073 (.073)	.483 (.430)	.757 (.430)
1B	SFT	.393 (.393)	.167 (.167)	.697 (.684)	.847 (.767)
1B	DPO	.550 (.510)	.380 (.380)	.933 (.873)	1.000 (.400)
1B	RLVR	.567 (.507)	.410 (.410)	.923 (.836)	1.000 (.440)
1B	Instruct	.590 (.550)	.363 (.356)	.900 (.820)	1.000 (.420)
7B	Base	.230 (.230)	.087 (.080)	.577 (.524)	.853 (.486)
7B	SFT	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	DPO	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	Instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 10: Personal-advice detector-positive rate, with the advice-minus-general- facts gap in parentheses ($\tau = 0.82$). Base uses plain QA; later checkpoints use chat serialization.

Model	Marker	Control	Base	SFT	DPO	Base→SFT	SFT→DPO
1B	referral	health facts	.188	.308	.167	+.121 (.414)	−.142 (.296)
1B	empathy	emotional	.017	.225	−.242	+.208 (.095)	−.467 (.001)
1B	structuring	neutral how-to	−.013	−.050	.000	−.037 (.653)	+.050 (.475)
1B	validation	health facts	.071	.146	.375	+.075 (.475)	+.229 (.095)
1B	validation	emotional	.021	.121	.225	+.100 (.293)	+.104 (.296)
1B	validation	neutral how-to	.058	.171	.350	+.113 (.189)	+.179 (.059)
7B	referral	health facts	.321	.067	.200	−.254 (.166)	+.133 (.293)
7B	empathy	emotional	.108	.467	−.175	+.358 (.016)	−.642 (.001)
7B	structuring	neutral how-to	.050	−.158	−.025	−.208 (.026)	+.133 (.095)
7B	validation	health facts	.092	.175	.317	+.083 (.475)	+.142 (.296)
7B	validation	emotional	.054	.125	.179	+.071 (.414)	+.054 (.537)
7B	validation	neutral how-to	.092	.163	.292	+.071 (.414)	+.129 (.189)

Table 11: Complete targeted-control checkpoint interactions at $\tau = .82$ (10k prompt-clustered permutations). Four of 24 transition tests have $q < .05$. Only the 7B empathy Base→SFT and SFT→DPO effects retain $q < .05$ at all three detector thresholds.

In the aggregate analysis, broad-referral differences are positive in eight of nine 1B sources (one tie) and all nine 7B sources; narrower-referral differences are positive in six and seven sources. After flexible length conditioning, adjusted ORs exceed one in eight of nine sources for broad referral and structuring at both sizes, and six of nine for the narrower set. Their ranges are broad referral .919–1.294 (1B) and .988–1.584 (7B), narrower referral .600–1.537 and .636–2.242, and structuring .718–1.213 and .718–1.324. Source intervals can be wide, so these are heterogeneity descriptions rather than source-level discoveries. The 19,890 7B rows with missing source labels are retained as the null-source stratum.

There are 367,082 distinct 1B prompt texts and 367,110 distinct 7B prompt texts; 21,551 1B rows and 21,574 7B rows belong to duplicated groups, with maximum multiplicity 15. Prompt-cluster covariance and one-pair-per-prompt estimates address this exact-text dependence. A separate content hash finds 21,508 distinct pairs shared across mixtures (21,515 multiplicity-matched rows), limiting independent-replication language.

I Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. To preserve dependence among sentences from the same response, we assign Dirichlet(1) weights to response keys in a 10,000-draw Bayesian bootstrap and recompute each metric. Table 15 reports percentile intervals. These intervals quantify sampling uncertainty around the AI consensus labels; they do not turn the audit into human validation.

J Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b + 1)/(B + 1)$. With $B = 10{,}000$, zero exceedances yield the minimum $p = 1/10{,}001 \approx 9.999 \times 10^{-5}$, not zero. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean seeded-marker per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs, Haldane–Anscombe matched odds ratios with log-scale intervals, paired risk

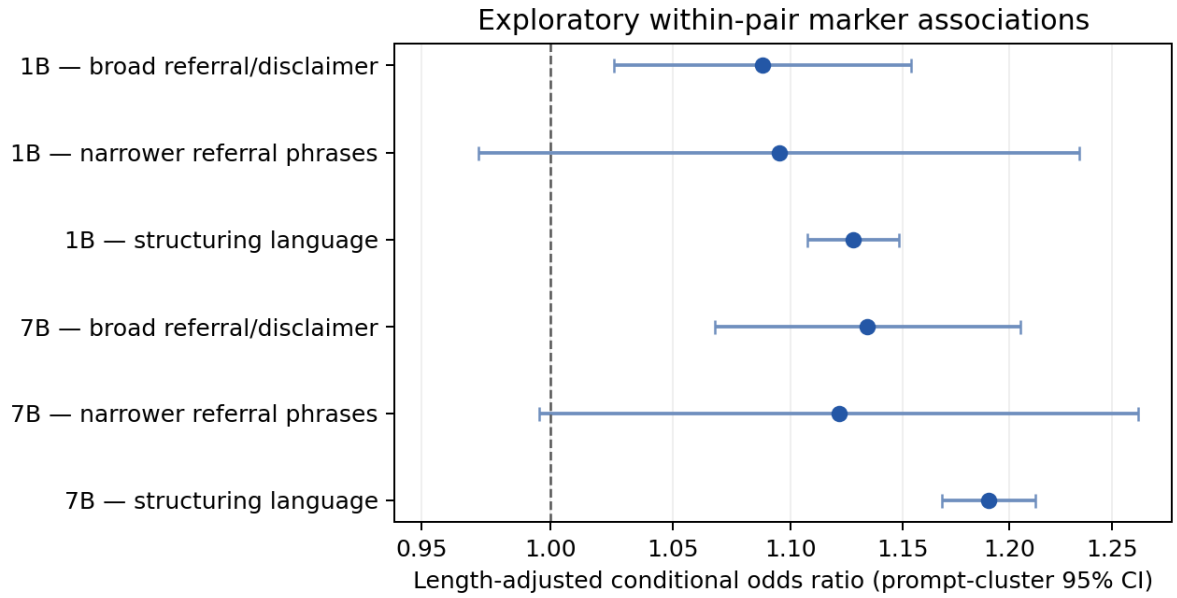


Figure 1: Exploratory paired conditional marker associations in the 1B/7B DPO training mixtures after flexible response-length modeling. Bars are 95% prompt-cluster sandwich intervals. OR $= 1$ denotes no modeled marker–choice association, not a zero change in preference probability.

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	consult a	891	978	1.10
1B	it's understandable	218	227	1.04
1B	it's natural to	152	157	1.03
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 12: Per-phrase SFT-assistant recoverability (representative rows). The 3 (1B) / 6 (7B) distinct 3-word types underlie the estimates in Appendix G; "here are some" dominates the 3-word mean.

differences with normal-approximation intervals, and phrase occurrences per million whitespace-delimited response words. The exploratory conditional fits use the no-intercept paired likelihood in Section 3.4, a five-knot restricted cubic spline of each side's log word count, and prompt-cluster sandwich covariance multiplied by $G/(G-1)\,(n-1)/(n-k)$. All aggregate and length-band counts were reproduced before fitting. A separate natural-spline implementation reproduced all six marker coefficients within 7.6×10^{-7} and their standard errors within 3.2×10^{-9}. The accompanying scripts derive all statistics from the artifact.

Mixture	Phrase group	Δ pp [95% CI]	MOR [95% CI]	Chosen only	Rejected only
1B	referral/disclaimer	.100 [.065,.135]	1.181 [1.114,1.252]	2478	2098
1B	narrower referral	.023 [.006,.040]	1.170 [1.039,1.318]	591	505
1B	structuring	1.393 [1.278,1.507]	1.242 [1.220,1.264]	27047	21779
7B	referral/disclaimer	.179 [.144,.214]	1.352 [1.274,1.434]	2602	1925
7B	narrower referral	.043 [.026,.061]	1.335 [1.187,1.502]	649	486
7B	structuring	2.280 [2.167,2.394]	1.435 [1.409,1.461]	28466	19838

Table 13: DPO paired absolute and relative effects. Positive Δ means more chosen-only than rejected-only matches.

Mixture	Word-difference band	n	Referral		Narrower referral		Structuring	
			Δ pp	MOR	Δ pp	MOR	Δ pp	MOR
1B	chosen longer > 20	171170	+.470	1.991	+.059	1.408	+5.979	2.361
1B	rejected longer > 20	134707	$-.310$	.616	$-.020$	.871	-3.786	.579
1B	within ± 20	72424	$-.010$	.965	+.017	1.238	+.184	1.062
7B	chosen longer > 20	187201	+.502	2.172	+.098	1.761	+6.562	2.610
7B	rejected longer > 20	120381	$-.223$	.706	$-.015$	.905	-3.239	.619
7B	within ± 20	70759	+.009	1.028	$-.003$	.966	+.343	1.123

Table 14: DPO effects stratified by chosen-minus-rejected response length. Positive Δ means more chosen-only matches.

Category	Precision 95%	Recall 95%	F1 95%
empathy expression	[.04,.46]	[.10,.90]	[.05,.57]
validation	[.02,.84]	[.03,.98]	[.02,.83]
referral	[.00,.22]	[1.00,1.00]	[.00,.36]
structuring	[.35,.63]	[.74,.96]	[.49,.74]

Table 15: Bayesian response-cluster bootstrap intervals for the 120-sentence AI consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.